

Explicit Semantic Modeling for Solving Word Association Puzzles

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Abstract

Word connection puzzles, such as the New York Times *Connections*, require partitioning a fixed set of words into disjoint subsets based on latent semantic, contextual, or structural relations. While large language models and embedding-based clustering methods show moderate success, prior analyses demonstrate they fail systematically on puzzles involving polysemy, semantic distractors, and form-based categories. This failure is largely due to their reliance on surface-level similarity and a lack of explicit relational constraint satisfaction. Cognitive and psycholinguistic research suggests that human word grouping relies on structured semantic networks rather than independent similarity judgments. This project investigates whether explicit semantic knowledge, combined with combinatorial scoring and global optimization, can better model the reasoning required for such puzzles. We propose a multi-signal similarity ensemble incorporating contrastive fine-tuning and beam search decoding, which successfully overcomes the critical limitations of traditional clustering and graph-based models.

1 Introduction

Word association and grouping tasks are fundamental to evaluating human language understanding and computational abstract reasoning. Games such as the New York Times *Connections* puzzle present a deceptively simple challenge: given a fixed board of 16 words (see Figure 1), the goal is to partition them into four groups of exactly four words, where each group shares a hidden relationship. These relationships may be semantic (e.g., *types of animals*), entity-based (e.g., *fictional universes*), or indirect and form-based (e.g., *phrase continuations* or *orthographic patterns*).

1.1 Motivation

While embedding-based similarity methods and large language models (LLMs) demonstrate partial success on general NLP tasks, they frequently fail on lateral thinking puzzles. Puzzles are deliberately adversarial; designers explicitly embed “red herrings”—words that are semantically close but belong to different groups.

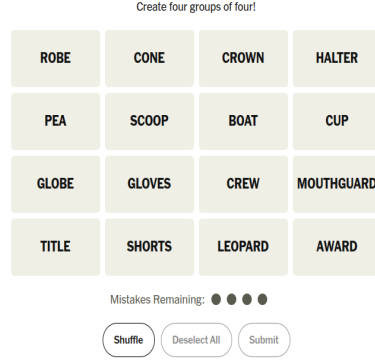


Figure 1: An example NYT *Connections* puzzle

Figure 2 illustrates how the correct groupings are revealed upon solving: colour-coded categories expose the hidden relational structure that surface-level similarity alone cannot recover. Furthermore, LLM-based approaches rely on opaque probabilistic inference, making it difficult to interpret, debug, or control their reasoning process. Addressing these shortcomings requires an architecture that moves beyond static representations and actively models the structural constraints of the game through explicit semantic modeling.

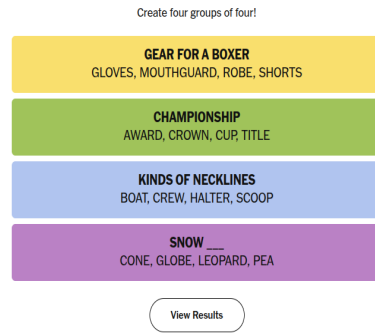


Figure 2: The solved state of the puzzle from Figure 1.

1.2 Objectives

This project is structured into three progressive components:

- **Part I:** Establish strong empirical baselines using embedding-based clustering and existing neural

approaches (GNNs) to mathematically quantify the task’s difficulty and identify primary failure modes.

- **Part II (Core Contribution):** Develop a deterministic, multi-signal similarity ensemble pipeline designed to capture diverse linguistic constraints (semantic, lexical, phonetic, and ontological) and solve the puzzles systematically using combinatorial optimization.
- **Part III (Extension):** Explore the automated generation of controlled puzzles, investigating whether learned structural constraints can be inverted to synthesize novel datasets via LLMs.

2 Literature Review

The *New York Times Connections* word game has rapidly evolved from a popular human puzzle into a rigorous benchmark for computational reasoning and constraint satisfaction. Recent literature has extensively documented the gap between human lateral thinking and current NLP capabilities, motivating the exploration of explicit semantic modeling.

2.1 Benchmarking Lateral Reasoning and Abstraction

Initial studies have established *Connections* as a stress test for Large Language Models (LLMs). **NYT-Connections** (2) introduced the game as a classification benchmark, demonstrating that even state-of-the-art models like GPT-4 fall significantly short of human performance when forced into isolated System-1 reasoning tasks. **Connecting the Dots** (6) analyzed over 400 puzzles, proving that multiple knowledge modalities—including encyclopedic, multiword, and form-based structural knowledge—are absolutely necessary for success. Furthermore, **Missed Connections** (10) framed the task as a direct measure of lateral thinking, highlighting how both embedding-based clustering baselines and zero-shot LLMs consistently fail without deeper relational constraint satisfaction. Conversely, work on **Puzzle Generation** (3) has demonstrated that models can successfully synthesize plausible puzzles using hierarchical prompting, indicating that the puzzle domain possesses a highly rigid, learnable structure.

2.2 Knowledge-Augmented and Contextual Semantic Representations

Solving lateral word association puzzles requires representations that capture both broad distributional context and explicit, real-world constraints. **MPNet**(8) addressed the limitations of previous contextual models with **MPNet**, which unifies Masked Language Modeling (MLM) and Permuted Language Modeling (PLM). By leveraging auxiliary position information, MPNet resolves the position discrepancy issues inherent in models like XLNet, yielding highly robust sentence-level embeddings.

However, for zero-shot lateral thinking tasks, purely distributional embeddings remain highly vulnerable to “red herring” semantic distractors. To mitigate this, **ConceptNet 5.5(9)** introduced **ConceptNet Numberbatch**, which injects structured, open-domain knowledge graphs into word embeddings. Using a retrofitting algorithm, Numberbatch forces distributional vectors to respect explicit relational constraints (e.g., *is-a*, *part-of*), proving that explicitly combining knowledge graphs with standard text representations significantly outperforms purely text-trained models on semantic relatedness evaluations.

2.3 Graph Neural Networks in Combinatorial Optimization

Initial computational approaches to modeling constraint-satisfaction tasks frequently utilize Graph Neural Networks (GNNs) to capture multi-hop lexical relationships. Recent work has formalized the theoretical capacity of GNNs in these domains. **Towards explaining the power of constant-depth graph neural networks for structured linear programming** (1) demonstrated that even constant-depth, shallow GNNs can effectively approximate solutions for structured linear programming, operating similarly to distributed optimization algorithms for localized, node-level predictions.

Despite strong node-level performance, extracting strict, globally valid partitions (such as exact sets of four disjoint cliques) from a continuous GNN output remains mathematically prohibitive. **Optimizing over trained GNNs via symmetry breaking**(11) highlighted that optimizing over trained GNNs requires framing the output through mixed-integer programming. Crucially, they identified that symmetric solutions—caused by graph isomorphism—create severe computational bottlenecks. They proposed necessary symmetry-breaking constraints to extract valid global optimums from GNNs, a requirement that significantly increases the computational overhead for lateral grouping tasks and often renders open-domain multi-hop traversal computationally infeasible.

3 Dataset Details and Augmentation

The input to the system is a raw string of 16 words. To train and evaluate the architecture, we require a robust, logically consistent dataset.

3.1 Primary Dataset

We utilize the official dataset of 914 historical NYT *Connections* games. To prevent temporal data leakage and ensure the model evaluates on genuinely unseen puzzles, the dataset is chronologically split into 814 training puzzles and 100 testing puzzles.

3.2 Data Augmentation Techniques

Because 814 training puzzles (yielding roughly 3,200 positive group examples) pose a severe risk of model

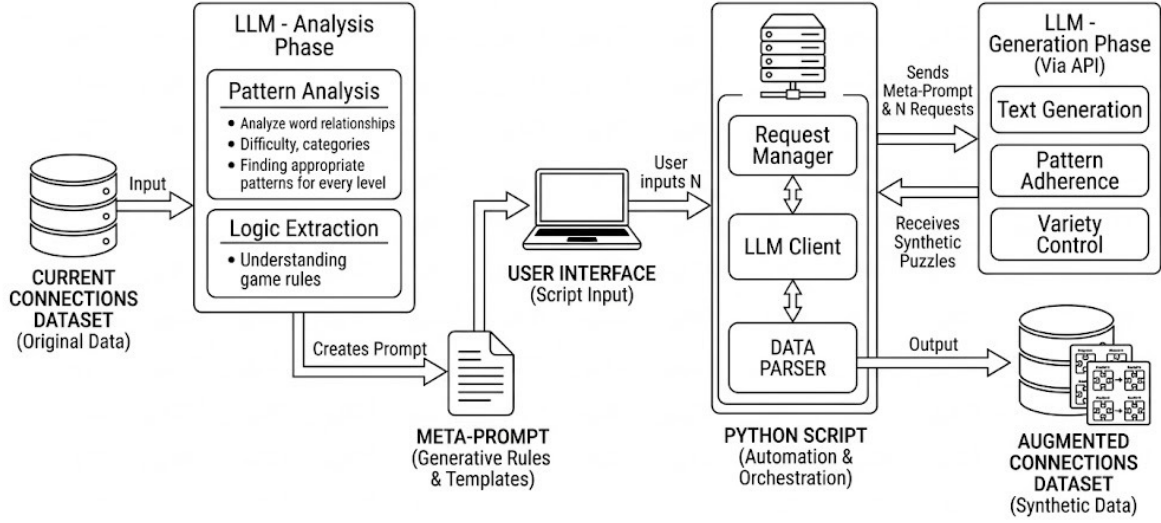


Figure 3: Automated LLM Data Augmentation Pipeline Architecture.

overfitting, we implemented two distinct data augmentation methodologies to expand the dataset without manual authoring.

3.2.1 Method 1: Automated LLM Data Augmentation Pipeline

To robustly expand the training corpus while preserving the nuanced lateral logic of the original game, we engineered an automated, two-phase LLM-driven augmentation pipeline, whose end-to-end architecture is illustrated in Figure 3.

- **Phase 1: Analytical Rule Extraction.** An initial LLM analyzes the historical dataset to deduce core game mechanics, difficulty gradients, and lateral wordplay patterns (e.g., intentionally embedded semantic distractors and overlapping categories). These codified constraints are synthesized into a strict, few-shot generative **Meta-Prompt**.
- **Phase 2: Automated Generation and Validation.** A Python-based orchestrator script systematically prompts a generative LLM via API using the Meta-Prompt. The orchestrator processes the raw text output, parses the puzzle structures, and enforces rigorous programmatic validation checks to ensure strict structural compliance (exactly four disjoint groups of four unique words).

This closed-loop system dynamically synthesizes a massive, logically consistent corpus of novel *Connections* puzzles without manual human authoring, directly mitigating the risk of downstream model overfitting on the limited baseline dataset. A representative example of a puzzle produced by this pipeline is shown in Figure 4.



Figure 4: A sample puzzle generated by Method 1

3.2.2 Method 2: Programmatic Knowledge-Graph Augmentation Pipeline

To complement the LLM-generated puzzles and ensure strict taxonomic accuracy, we engineered a deterministic, programmatic puzzle generator whose architecture is depicted in Figure 5. This system synthesizes mathematically validated puzzles by querying four distinct linguistic knowledge bases:

- **BabelNet (Word Sense Disambiguation):** Maps anchor concepts to precise WordNet offsets, preventing semantic bleed (e.g., ensuring “apple” maps strictly to the fruit synset rather than the technology company) before expansion.
- **WordNet (Lexical Structure):** Supplies strict semantic relations, extracting hyponym trees (for

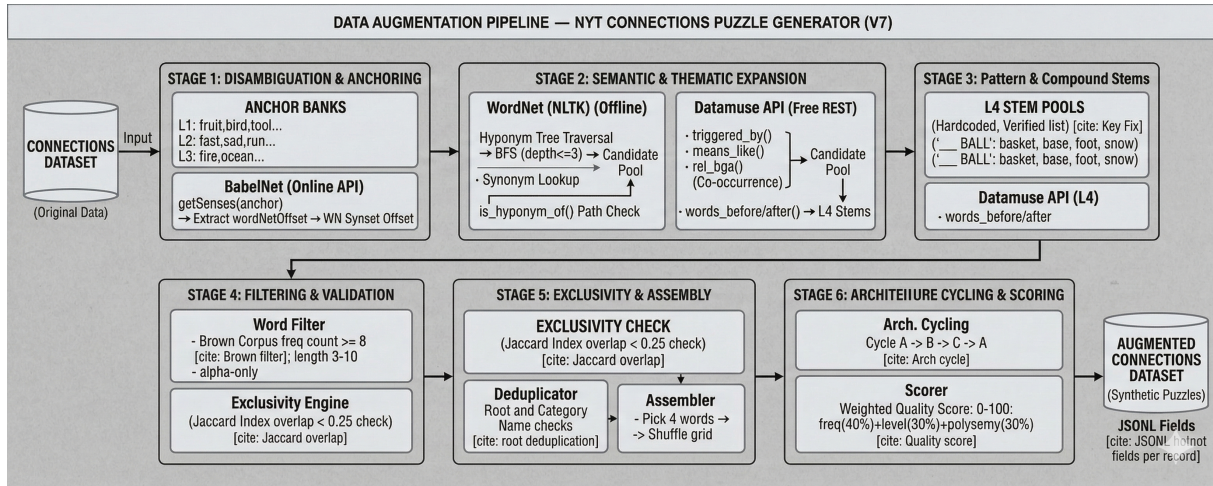


Figure 5: Programmatic Knowledge-Graph Augmentation Pipeline Architecture

concrete categories) and meronym relations (for part-whole categories).

- **Datamuse API (Thematic & Structural):** Retrieves soft, cultural associations (e.g., “words associated with fire”) and compound word components (e.g., prefixes/suffixes) that strict dictionary taxonomies lack.
- **Brown Corpus (Linguistic Filtering):** Enforces a minimum frequency threshold (e.g., $\text{freq} \geq 8$) to guarantee all generated candidates are commonly known English vocabulary, rejecting obscure taxonomic terms.

Workflow and Combinatorial Constraints. The pipeline constructs four distinct difficulty tiers: L1 (Concrete Categories), L2 (Synonym Groups), L3 (Thematic Associations), and L4 (Compound Word Stems). To ensure puzzle validity, the system enforces several strict quality controls:

- **Semantic Exclusivity (Jaccard Filtering):** For L1, L2, and L3, candidate words are cross-referenced against all other groups using Jaccard synset overlap. Candidates exceeding a 0.25 overlap threshold are rejected to prevent unintentional ambiguity.
- **Engineered Misdirection (L4 Exemption):** Crucially, L4 (Compound stems) is deliberately exempted from the Jaccard exclusivity check. Allowing L4 component words (e.g., “foot”, “fire”) to semantically overlap with L1–L3 categories mathematically replicates the intentional “red herring” design of human-authored puzzles.
- **Architectural Cycling:** To prevent the downstream NLP model from overfitting to a single generative pattern, the pipeline sequentially cycles through three distinct puzzle architectures (Standard Hyponyms, Abstract/Contrast-Based, and

Meronym-Based), maximizing structural and vocabulary diversity across the generated JSONL dataset.

A representative puzzle produced by Method 2 is shown in Figure 6, demonstrating the taxonomic precision and controlled misdirection enforced by the knowledge-graph pipeline.

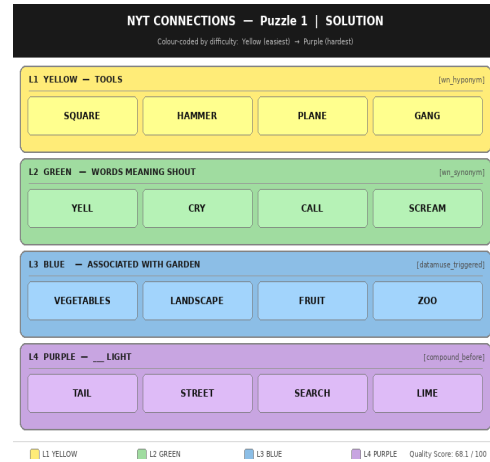


Figure 6: A sample puzzle generated by Method 2

4 Methodology

Because the puzzle intentionally utilizes “red herrings,” standard zero-shot clustering models fail. To solve this problem, we propose an advanced **Multi-Signal Similarity Ensemble** enhanced by contrastive fine-tuning, extended feature extraction, and beam search decoding. The full architecture is shown in Figure 7.

4.1 Multi-Modal Word Representation

Each word is encoded using three distinct embedding strategies:

- **Contextual Semantics:** A dense sentence transformer (MPNet or E5-Large) is **contrastively fine-**

tuned on the training set using a triplet loss function. This adapts the space directly for the *Connections* task, pulling true group members together and pushing distractors apart.

- **Distributional Semantics:** GloVe (100-dim) captures static, topic-level co-occurrence statistics.
- **Ontological Semantics:** ConceptNet Numberbatch (300-dim) captures explicit, common-sense world knowledge.
- **Joint Embedding:** The fine-tuned contextual vector is concatenated with the Numberbatch vector and L_2 -normalized, yielding a 1,068-dimensional representation that merges task-optimized semantics with structured graph knowledge.

4.2 Pairwise Similarity Computation & Normalization

We compute nine independent 16×16 similarity matrices, each measuring a specific relational signal between all word pairs: Semantic (MPNet), Lexical (Character n-gram), WordNet (Wu-Palmer), WikiData (Entity overlap), Datamuse, ConceptNet, Wikipedia Categories, Phonetic (CMU Edit Distance), and Morphological (Affix overlap).

To eliminate “hubness” (systematic bias from words universally similar to others), each matrix is independently double-centered before fusion:

$$\tilde{S}(i, j) = S(i, j) - \mu_{\text{row}}(i) - \mu_{\text{col}}(j) + \mu_{\text{global}} \quad (1)$$

4.3 Downstream Optimization & Beam Search Decoding

As shown in Figure 7, the architecture branches into two parallel solvers to satisfy the strict global game constraints (exactly 4 groups of 4). Both paths replace standard greedy decoding with **Beam Search** ($K = 5$) with a two-step lookahead to prevent early commitment errors.

- **Path A (Deterministic Inference):** Aggregates the nine normalized matrices via a linear combination. Weights are learned using Logistic Regression on the validation set. An Uncertainty Filter is applied to skip candidate partitions where model agreement is low.
- **Path B (Machine Learning Ensemble):** Extracts approximately 60 features per candidate 4-word group, replacing previous aggregate statistics with raw pairwise scores, intra-group rank features, and cross-matrix consistency metrics (Pearson correlation across the 9 signals). A Random Forest and Gradient Boosting ensemble is trained on hard negatives (groups with exactly 3 correct words) to strictly classify valid partitions.

5 Progress So Far: Architectural Evolution & Completed Objectives

To mathematically quantify the difficulty of the task and isolate necessary architectural features, several baseline models were developed, evaluated, and documented. These completed objectives heavily informed the proposed methodology above.

5.1 Objective 1: Unsupervised Clustering Baselines (Completed)

- **Baseline 0 (GloVe + K-Means):** Mapped words using static GloVe 300d vectors and optimized via K-Means and the Hungarian Algorithm. Failed due to an inability to disambiguate polysemous words.
- **Baseline 1 (GloVe + SBERT Agglomerative):** Concatenated GloVe and all-MiniLM-L6-v2 contextual vectors, clustering via greedy agglomerative distance. Proved highly vulnerable to the “Red Herring” trap; greedy clustering made irreversible groupings based on immediate similarity, rendering it physically incapable of backtracking.
- **Baseline 2 (MPNet + Double Centering):** Utilized all-mpnet-base-v2 combined with lexical similarity and a Double Centering transformer. Achieved an 11% game solve rate but suffered cascade failures on abstract trivia groups.

5.2 Objective 2: Graph Neural Network Exploration (Completed)

To mathematically capture the multi-faceted, lateral relationships between words, we formulated the 16-word board as a heterogeneous multigraph. The target words served as nodes, while semantic, lexical, and entity-based relationships served as distinct edge types. We developed and evaluated two separate Graph Neural Network (GNN) paradigms. Both models ultimately collapsed, yielding a 0% full-game solve rate, which empirically exposed the limitations of deep graph learning for this specific constraint-satisfaction domain.

5.2.1 Localized Node Optimization (Node Classification)

The first iteration formulated the puzzle as a supervised, localized node classification problem.

Architecture: The model utilized a Relational Graph Convolutional Network (R-GCN) featuring distinct learnable weight matrices for specific edge types (e.g., WordNet taxonomic links versus orthographic prefix matches). After aggregating neighborhood features via heterogeneous message passing, a softmax classification head attempted to predict the exact group assignment (Class 0, 1, 2, or 3) for each of the 16 isolated nodes.

Failure Mode (The Arbitrary Label Contradiction): This formulation suffered from a fundamental theoretical flaw. The group IDs (0, 1, 2, 3) are arbitrary and highly puzzle-specific; Class 0 might represent the semantic category “Wet Weather” in one puzzle and the

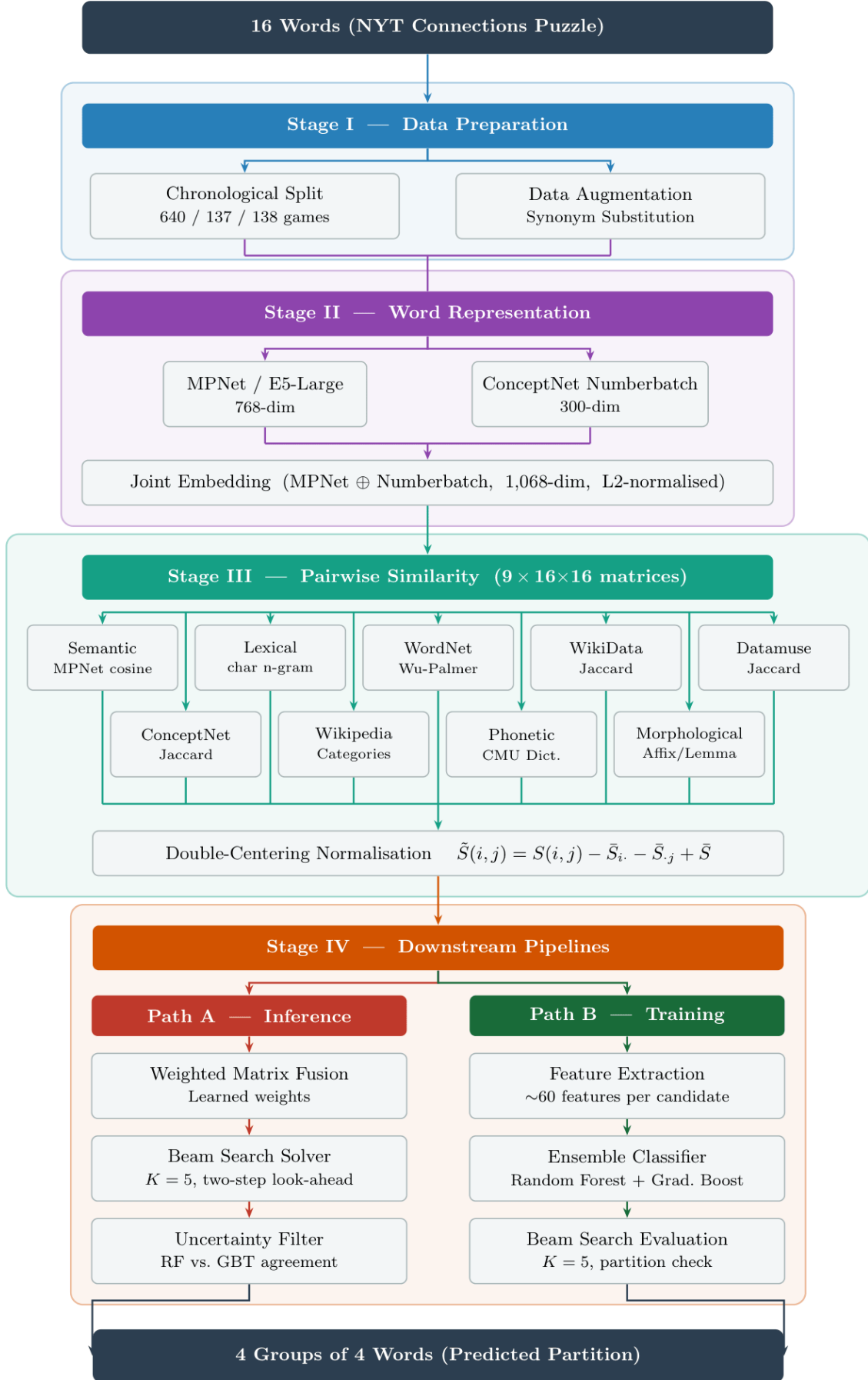


Figure 7: Architecture diagram of the proposed Multi-Signal Similarity Ensemble.

structural category “Palindromes” in the next. The network could not mathematically learn globally consistent weights for dynamically shifting, context-dependent labels. Furthermore, localized node classification failed to enforce overarching game rules, frequently predicting mathematically invalid board states (e.g., assigning five words to a single group).

5.2.2 Global Optimization (Partition Scoring Model)

To resolve the structural violations of the node classifier, the architecture was pivoted to evaluate the board holistically via pairwise link prediction. The R-GCN was tasked with outputting a continuous probability matrix for all 120 possible word pairs ($16 \times 15/2$). To stabilize this architecture, several advanced engineering interventions were implemented:

- **Offline Caching:** Live API calls were replaced with 300-dimensional ConceptNet Numberbatch pre-trained embeddings to prevent prohibitive I/O bottlenecks during the training loop.
- **Symmetric Feature Extraction:** Standard pair concatenation was replaced with the absolute difference and element-wise product of embeddings, explicitly providing the network with invariant semantic distances to score.
- **Loss Weighting & Skip-Connections:** To handle extreme class imbalance (24 positive vs. 96 negative edges per board), a `pos_weight` of 4.0 was injected into the Binary Cross Entropy loss function. Additionally, original Numberbatch embeddings were wired directly to the final Multi-Layer Perceptron (MLP) scorer via skip-connections to preserve baseline semantics through dead graph convolutions.

5.2.3 Core Drivers of Architectural Collapse

Despite these heavy structural interventions, the global optimization GNN severely underfit the data, stalling at a uniform loss plateau. The collapse was driven by three compounding factors:

1. **The “Empty Graph Collapse” (Extreme Sparsity):** *Connections* heavily utilizes lateral word-play (e.g., shared prefixes, homophones) rather than strict ontological links. Consequently, graphs generated from explicit knowledge bases were incredibly sparse. In a GNN, nodes without explicit edges receive no message-passing updates. This effectively zeroed out their embeddings during convolutions, blinding the network to isolated words.
2. **Noise and Relation Overload:** Attempting to build graphs from expansive open-domain knowledge bases like WikiData introduced an overwhelming number of distinct relation types. This massive variance injected noise into the graph,

making it mathematically impossible for the shallow R-GCN to distinguish between highly relevant connections and coincidental, irrelevant associations.

3. **Data Insufficiency:** Graph Neural Networks are inherently data-hungry. The dataset of 814 training puzzles (yielding roughly $\sim 97,000$ word pairs) was insufficient to effectively generalize the complex, multi-relational message-passing weights required to navigate disparate edge types.

5.3 Objective 3: Initial Ensemble Inference (Completed)

To bypass the limitations of greedy clustering and GNN sparsity, we engineered a deterministic, multi-modal ensembling pipeline. This prototype extracted four orthogonal 16×16 similarity matrices: Semantic (MP-Net), Lexical (n-grams), Ontological (WordNet), and Entity-Based (WikiData). To prevent high-dimensional “hubness,” each matrix was normalized via Double Centering and fused into a unified matrix. Framed as a combinatorial optimization problem, the solver evaluated all $\binom{16}{4}$ possible disjoint partitions. Achieving a 16% full-game solve rate, this validated our hypothesis: fusing heterogeneous linguistic signals strictly outperforms single-vector approaches, establishing the foundation for the advanced architecture in Figure 7.

6 Evaluation Strategies

The models are evaluated against the unseen 100-puzzle test set using strict combinatorial metrics:

1. **Game Accuracy:** Total number of full games solved completely (4 out of 4 groups perfectly partitioned).
2. **Grouping Accuracy:** The total absolute number of individual groups solved (out of 400 possible groups).
3. **The “One Away” Metric:** The number of groups where exactly 3 out of 4 words were correctly clustered (indicating the model found the correct latent relationship but failed final disambiguation).
4. **Search Efficiency:** The number of guesses (attempts) required by the beam search algorithm to output the final board state.

Table 1 summarises the comparative performance of all architectures on the held-out test set. The progressive improvement from Baseline 0 through Ensemble-Itr-1 directly reflects each architectural decision described in Section 5.

7 Tentative Timeline and Future Work

The project is currently transitioning from the successful validation of the Ensemble-Itr-1 architecture (Table 1) into the advanced methodology proposed in Section 4 and depicted in Figure 7.

Architecture / Model	Games Solved	Groups Solved	Groups “One Away”	Games “One Away”
Baseline 0 (GloVe + K-Means)	1% (1/100)	10.50% (42/400)	33.25% (133/400)	8% (8/100)
Baseline 1 (GloVe + SBERT Agglo.)	0% (0/100)	22.50% (90/400)	19.75% (79/400)	0% (0/100)
Baseline 2 (MPNet + Double Center.)	11% (11/100)	24.25% (97/400)	43.25% (173/400)	3% (3/100)
GNN (Node / Global Opt.)	0% (0/100)	3.00% (12/400)	19.00% (76/400)	0% (0/100)
Ensemble-Itr-1	16% (16/100)	33.50% (134/400)	43.50% (174/400)	10% (10/100)

Table 1: Comparative evaluation of all architectures on the 100-puzzle held-out test set.

- **Ongoing (March 2026):** Implementing “Ensemble-Itr-2”. This involves expanding from 4 to 9 similarity signals (adding phonetic and morphological features) and integrating the Random Forest/Gradient Boosting classifiers to replace the greedy combinatorial search with Beam Search decoding. Execution of the Contrastive Fine-Tuning loop for the MPNet/E5-Large embeddings using the triplet loss function on the training dataset is also underway.
- **Planned (April 2026):** Refinement of the post-hoc reasoning LLM generator (flan-t5-base) to act as a Neuro-Symbolic explainer, ensuring textual output perfectly aligns with the underlying matrix mathematics. Final evaluation, ablation studies, and completion of Part III (Optional LLM puzzle generation).

8 Conclusion

This work systematically investigates the challenge of solving the New York Times *Connections* puzzle through explicit semantic modeling. Using the puzzle and solution examples in Figures 1 and 2 as motivation, we established that standard clustering and graph-based methods fail due to polysemy, semantic distractors, and the sparse relational structure inherent to lateral wordplay tasks. Through iterative architectural development—from unsupervised GloVe baselines to GNN exploration to multi-signal ensemble inference—we identified the critical role of heterogeneous signal fusion and combinatorial global optimization. Our Ensemble-Itr-1 achieves a 16% full-game solve rate (Table 1), validating the multi-signal approach and providing a strong foundation for our proposed advanced architecture (Figure 7). The data augmentation pipelines depicted in Figures 3 and 5 further support scalable training by producing structurally valid, diverse puzzle corpora. The integration of contrastive fine-tuning, beam search decoding, and a broader nine-signal similarity ensemble represents a principled path toward closing the remaining gap with human-level performance on this benchmark.

9 GitHub Repository

The full project codebase, including all augmentation pipelines, baseline implementations, and ensemble architectures, is publicly available at: https://github.com/Anurag-Kacholiya/NYT-Connection_Solver

References

- [1] Chen, Z., et al. (2024). Towards explaining the power of constant-depth graph neural networks for structured linear programming. In *ICLR 2024*.
- [2] Lopez, A. Y. L., McDonald, T., & Emami, A. (2024). NYT-Connections: A deceptively simple text classification task that stumps system-1 thinkers. *arXiv:2412.01621*.
- [3] Merino, T., et al. (2024). Making new connections: LLMs as puzzle generators for the New York Times’ Connections word game. *arXiv:2407.11240*.
- [4] Mikolov, T., et al. (2013). Efficient estimation of word representations in vector space. *arXiv:1301.3781*.
- [5] Miller, G. A. (1995). WordNet: A lexical database for English. *Communications of the ACM*, 38(11):39–41.
- [6] Samadarshi, P., et al. (2024). Connecting the dots: Evaluating abstract reasoning capabilities of LLMs using the New York Times Connections word game. *arXiv:2406.11012*.
- [7] Schlichtkrull, M., et al. (2018). Modeling relational data with graph convolutional networks. In *European Semantic Web Conference*.
- [8] Song, K., et al. (2020). MPNet: Masked and permuted pre-training for language understanding. *arXiv:2004.09297*.
- [9] Speer, R., Chin, J., & Havasi, C. (2017). ConceptNet 5.5: An open multilingual graph of general knowledge. *arXiv:1612.03975*.
- [10] Todd, G., et al. (2024). Missed connections: Lateral thinking puzzles for large language models. *arXiv:2404.11730*.
- [11] Zhang, S., et al. (2023). Optimizing over trained GNNs via symmetry breaking. *arXiv:2305.09420*.